Behavioral and Immunological Consequences of Brief Mother-Infant Separation: A Species Comparison

M. L. LAUDENSLAGER
P. E. HELD
M. L. BOCCIA
M. L. REITE
J. J. COHEN
University of Colorado Health Sciences Center,
Department of Psychiatry
Denver, Colorado

The immediate behavioral and immunological consequences of a single 2-week maternal separation experience were studied in socially housed 7-month old bonnet and pigtail macaques. Maternal separation was associated with species dependent behavioral changes. Both species showed significant increases in ingestive behaviors associated with separation. Separated and matched controls showed an increase in disturbance behaviors (vocalization, startles, shaking, temper-tantrums) that subsided after 24–36 hours in control subjects, but continued, albeit with species-dependent patterns, in the separated monkeys. Allomaternal care of the separated bonnet infants but not the pigtail infants was associated with a progressive reduction of these disturbance behaviors during the separation period. Following a period of agitation, pigtail infants showed a depressive phase characterized by slouched, withdrawn postures and reduced motor activity. As a group, the separated infants of both species were not different from controls with respect to lymphocyte activation by mitogens, a measure of immunocompetence. However, when individual behavioral responses were considered, the change in lymphocyte activation during separation was significantly related to behavioral responses which reflected disturbance, such that the change in lymphocyte activation following *in vitro* stimulation with the mitogens phytohemagglutinin and Concanavalin A (markers of the immunocompetence of T lymphocytes) was related to levels of vocalization and time spent in slouched postures. The activation of B lymphocytes by pokeweed mitogen was not influenced by the separation experience nor was it associated with specific behavioral responses to separation. The importance of assessing the affective consequences of stressors is discussed.

Reprint requests should be sent to Mark L. Laudenslager, University of Colorado Health Sciences Center, Department of Psychiatry, 700 Delaware Street, Denver, CO 80204.

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Introduction

The behavioral impact of parental separation or loss in young children may be quite profound (Bowlby, 1960). Recent loss experiences have been associated with increased morbidity and mortality among the survivors (Laudenslager & Reite, 1984; Stroebe & Stroebe, 1987). The magnitude of the association found in epidemiological studies between social stressors and health, however, is frequently small and reaches statistical significance due to the statistical power of the large number of subjects typically studied (Rabkin & Strueling, 1976). Furthermore, epidemiological studies rarely assess the affective consequences of these stressors, resulting in a heterogeneous bereavement group. For example, Wortman and Silver (1987) described individuals who had experienced a recent loss and yet showed minimal or absent bereavement responses. Although the grief response has generally been viewed as a normal process following the loss of a significant other, Wortman and Silver suspect that individuals lacking a profound response to loss may be showing resiliency rather than psychopathology. Others have also suggested that the small correlations between significant life events and negative health consequences may be due, in part, to individual variation in the response to the stressors. Kobasa's (1979) concept of "hardiness" suggested that certain individuals possess a number of positive personality traits that result in resistance to stressors. The concept of hardiness is not without its problems, but such a theoretical perspective emphasizes the need for detailed assessment of individual responses to presumed stressful life events.

Animal models have indicated that knowledge of behavioral responses following stressor exposure permits accounting for a greater proportion of the variance in biological responses occurring at the time of the stressor experience. In rodents, for example, characterization of the behavioral response of a colony intruder to the residents accounted for as much as 46% of the variance in serum antibody levels produced in response to an antigen administered at the time of group introduction (Fleshner, Laudenslager, Simons, & Maier, 1989). It is, thus, critical to characterize the behavioral component following stressful experiences in attempting to relate specific stressors to biological responses. The relatively small effects and high level of variability observed when subjects are grouped, based on exposure to a common stressor, may be related to heterogenous behavioral responses to the stressor.

The consequences of recent loss or separation experiences in adult populations include changes in components of the immune system, such as *in vitro* responses of lymphocytes to mitogens and natural cytotoxicity toward tumor cells (Bartrop, Luckhurst, Lazarus, Kiloh, & Penny, 1977; Calabrese, Kling, & Gold, 1987; Irwin, Daniels, Risch, Bloom, & Wiener, 1988; Schleifer, Keller, Camerino, Thornton, & Stein, 1983; Kiecolt-Glaser, Fisher, Ogrocki, Stout, Speicher, & Glaser, 1987. Epidemiological studies have documented increased risk for diseases of nonimmunological origin, such as cardiovascular disease (Stroebe & Stroebe, 1987). Limited evidence would suggest that for adult populations, infectious disease and autoimmune processes, such as arthritis, are exacerbated in association with recent loss (see Laudenslager & Reite, 1984 for a review). In general, the underlying biological changes associated with loss, such as altered immune regulation, are poorly understood, particularly for humans.

It has been argued that many of the biological changes associated with loss in

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young animals are related to the role the mother plays as a regulator for a number of physiological systems (Hofer, 1984). Animal models often provide unique insights into problems otherwise not obvious in the study of human populations. Many of the components of the grief response following loss are also seen in association with a brief maternal separation experience in nonhuman primates (Mineka & Suomi, 1978) and have been proposed as an adequate animal model of grief (McKinney & Bunney, 1969). Herein, the relationship between the behavioral response to brief maternal separation and an *in vitro* measure of immune regulation, lymphocyte activation by mitogens, were investigated in bonnet (*Macaca radiata*) and pigtail (*M. nemestrina*) monkeys. These two species represent distinctively different patterns of maternal care in the social groups. Pigtail monkeys show relatively infrequent allomaternal care, whereas bonnet monkeys demonstrate considerably higher levels of allomaternal care (for reviews see Quiatt, 1979; Thierry & Anderson, 1986). Herein, we consider the presence of allomaternal care, as seen in the bonnet monkeys, to represent an animal model of social support. The role of social support in moderating behavioral and biological responses to stressful events (Cohen, 1988; House, Landis, & Umberson, 1988) has received considerable attention in human studies, but animal models of social support have been rare or nonexistent. Matched controls were also studied concurrently. These subjects served as controls for handling and group disturbance, an important control frequently omitted in many previous studies. This is particularly important in light of the impact of handling on the immune response in nonhuman primates (Hill, Greer, & Felsenfeld, 1967).

Methods

Subjects

A total of 21 pigtail (*Macaca nemestrina*) and 16 bonnet (*M. radiata*) monkey infants, ranging in age from 119 to 375 days at the beginning of the study, served as subjects. All subjects were reared by their mothers in social groups consisting of an adult male, 6–8 adult females, and a number of juveniles. Each separated infant ($N = 12$ pigtails, $N = 8$ bonnets) was studied with a concurrent control subject ($N = 9$ pigtails, $N = 8$ bonnets) of approximately the same age and sex. Descriptions of control and separated pigtail and bonnet monkey pairs are listed in Tables 1 and 2. On occasion, an older infant served as a control when there were no younger subjects present in the social group. Housing consisted of 2.1 × 2.5 × 4 M pens with glazed cinderblock walls, shelves for sitting, and PVC pipes for climbing. Subjects were observed, unobtrusively, through a one-way mirror from outside the pen. Lights were on between 0700 and 2000 hrs. Feeding occurred at 0900 and consisted of standard monkey chow and fresh fruit, which was spread uniformly across the floor of the pen. Water was available *ad lib* through a water spout in the pen.

Procedures

Behavioral observations were conducted using 5 min focal animal samples twice daily for all phases of the experiment, in the morning (between 1000 and 1200 hrs) and afternoon (between 1300 and 1600 hrs), 5 days per week. On the day of separation and the day of reunion, 6 behavioral samples were collected, 3 in the morning and 3 in the afternoon. Observations were collected in real time on a

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Table 1
Pigtail Monkey Separated/Control Pairs.

Subject Number	Condition	Social Group	Ma's Rank	Age at Separation	Sex	Immune Panel Included
32.5	Control	B	—	236	F	—
53.10	Separated[2]	B	—	209	M	—
63.2.3	Separated[2]	B	—	171	F	—
7.2.4.2	Control	G	7	294	F	—
8.5.3	Separated[2,3]	G	5	238	F	—
63.8	Separated[2]	G	2	226	M	—
49.11	Control	G	1	276	F	X
16.7.2	Separated	G	6	232	M	X
36.6.2	Control	G	8	158	F	X
42.3.3	Separated	G	9	166	M	X
72.4.4	Control	N	4	260	F	X
36.6.3	Separated	N	7	206	F	X
42.3.4	Control	N	5	175/232	M	X
8.5.5	Separated[1]	N	1	169	M	X
83.1.4	Separated[1]	N	2	246	F	X
49.2.4	Control	S	5	393	F	X
34.1.5	Separated	S	4	239	F	X
17.2.7	Control	S	2	295	F	X
17.3.6	Separated	S	4	238	M	X
17.3.5	Control	S	3	151	M	X
34.1.7	Separated	S	4	174	F	X

[1] Sequential separation.
[2] Simultaneous separation.
[3] Three week separation.

PDP 11/23 computer using a behavioral taxonomy developed by Kaufman and Rosenblum (1966) as modified previously in our laboratory (Reite & Short, 1983) and, more recently, as indicated below.

The separation occurred at approximately 7 months of age (mean age of separation = 209 days, see Tables 1 and 2) following a 2-week period of observation of the infant and mother in their social group (baseline). Both control and experimental subjects were observed during these sessions. Separations and reunions occurred on Monday morning following blood sampling. On the morning of separation, the mother and infant were moved from the social group at 0800 hr into a transport cage from which the infant was removed. The infant was then returned to the social group to begin the separation period, which lasted 2 weeks. Observations began immediately. The mother was housed individually in another part of the laboratory out of sight, sound, and smell of the infant. On the day of reunion (Monday), the mother was returned to the social group at 0900 hr and behavioral observations began immediately. Behavioral observations were continued for 2 subsequent weeks (reunion period).

Blood was obtained, for immunological evaluation, by venipuncture from both

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Table 2
Bonnet Monkey Separated/Control Pairs.

Subject Number	Condition	Social Group	Ma's Rank	Age at Separation	Sex	Immune Panel Included
2.2.7	Control	A	3	161	M	X
23.1.5	Control	A	1	169	F	X
5.1.5	Separated	A	2	161	M	X
23.1.2.1	Control	A	2	224	F	X
23.1.6	Separated	A	1	209	M	X
2.9.1	Control	C	—	349	M	—
6.2.5.3	Separated[1]	C	—	170	F	—
2.8.3	Separated[1]	C	—	211	M	—
2.6.11	Control	R	4	169	M	X
2.7.6	Control	R	5	133	F	X
3.4.15	Separated	R	1	152	F	X
2.2.1.10	Control	R	1	286	F	X
2.6.12	Separated	R	4	221	F	X
7.2.4	Control	T	—	265	M	—
3.8.5	Separated[1]	T	—	195	F	—
2.2.9	Separated[1]	T	—	342	M	—

[1] Simultaneous separation.

the control and experimental subjects at 0700 hr on Monday and Friday of each week throughout the period of behavioral observations, except as noted above. In a limited number of subjects tested in the latter phases of this project, blood was obtained from both control and separated subjects 2 hr following separation to determine the acute changes that might occur during the early agitation period. Subjects, accompanied by their mother except during the separation phase, were briefly isolated from the social group and highly restrained in a transport cage while 2–3 ml of blood were obtained. The entire procedure, from the time the experimenter entered the housing pen until the blood was drawn, was completed within 15–20 min for both subjects.

After blood was obtained from the subjects, 100 μl was removed for a differential cell count and the determination of total white blood cell count (WBC), which was performed on a Coulter Counter (Model ZBI). The remainder of the sample was overlaid on 3 ml of Ficoll Paque (Pharmacia) for isolation of lymphocytes cells by standard procedures reported elsewhere (Laudenslager, Harbeck, & Reite, 1982). In brief, recovered lymphocytes were washed three times in Dulbecco's phosphate buffered saline (DPBS) and then resuspended to 1×10^6 cells/ml of supplemented (l-glutamine, penicillin/streptomycin, 5% fetal calf serum) media RPMI-1640. Cells (1×10^5) were stimulated in triplicate with phytohemagglutinin (PHA), concanavalin A (ConA), and pokeweed mitogens (PWM) at several different concentrations (PHA: 0.5, 0.25, and 0.1 μg/well, ConA: 1.0, 0.75, 0.5, and 0.25 μg/well, and PWM: 3.0, 2.0, and 1.0 μg/well). After incubation at 37°C in a humidified CO_2 incubator for 72 hours, cells were pulsed with 1 μCi of H^3-thymidine and harvested 4 hours later onto glass wool filter discs. Filter discs were counted on

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Table 3
Behavioral Taxonomy System as Used in the Present Study.

A. *Activity/Environmentally Directed Behaviors*
   1. Locomotion[1]—Whole body movement greater than 15 cm
   2. Rest[1,3]—state description: subject quiet, either awake or asleep
   3. Exploration[1,3]—object or bedding exploration either manually or orally, active visual searching outside the pen
B. *Self-Maintenance Behaviors*
   1. Self-Groom[1]—animal actively picks at or brushes its fur.
   2. Ingest[1,3]—includes any eating and/or drinking behaviors
C. *Disturbance Behaviors*
   1. Vocalization[2,3]—coo, squeal, and gecker
   2. Oral self[1]—emphasis on sucking of self, such as fingers or other appendage
   3. Slouch[1]—a postural collapse of a sitting animal, characterized by a lack of motor tonus
   4. Stereotypic motor response[1]—unusual repetitive behavior patterns
   5. Disturbance[2,3]—Startle responses, "wet dog shake", scratch self, temper tantrums
D. *Non-maternal Social Behaviors*
   1. Play[2,3]—active mutual play with another animal, initiate play with another animal, receive play from another animal
   2. Groom[1]—infant grooms another animal
   3. Receive groom[1]—infant is groomed by another member of the social group
   4. Ventral/ventral[1,3]—cradled or enclosed by another member social group
   5. Near[1]—infant less than 60 cm with or without contacting another group member
   6. Agonistic[2,3]—initiate threat, receive threat, submit to another member of the social group, initiate aggression, receive aggression
E. *Maternal Social Behaviors*
   1. Ventral/ventral[1,3]—cradled by mother, enclosed by mother, and/or nipple contact by the infant
   2. Touch[1,3]—in contact with mother or passively supported by mother
   3. Affiliative (Ma)[1,3]—infant follows mother, infant initiates social exploration of mother, infant receives social exploration from mother, infant glances toward mother, infant orients rump toward mother with tail raised
   4. Away[1,3]—infant apart from its mother either on the same or another level of the pen
   5. Wean[2,3]—mother removes nipple from infant's mouth, mother denies nipple contact, infant receives punishment from mother, mother defers contact from the infant

[1] Duration category.
[2] Frequency category.
[3] Collapsed category.

a liquid scintillation counter. In order to minimize variations between technicians in the mitogen assay (see Maier & Laudenslager, 1988), we analyzed only the results generated by one of the authors (PEH) to minimize intertechnician error. This reduced the sample size to a total of 25 subjects (12 separated and 13 control subjects, 10 bonnets and 15 pigtail monkeys) for the immunological portion of the study. Pairings of subjects included in this analysis are indicated in Tables 1 and 2.

Data Analysis

The taxonomy used in the present study for the collection of behavioral observations was a subset of the laboratory's taxonomy (Kaufman & Rosenblum, 1966) which contains over 120 individual behaviors (including frequency and duration behaviors). Even the smaller subset of behaviors required abbreviation prior to analysis of the present data. A number of behavioral categories were collapsed

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into larger categories based on theoretical as well as correlational and cluster analyses. The resulting behaviors fell into 5 major categories: activity/environmentally directed behaviors, self maintenance behaviors, disturbance behaviors, nonmaternal social behaviors, and maternal social behaviors (see Table 3). Behaviors observed in baseline, separation, or reunion phases for less than 50% of the animals, on the basis of infrequent occurrence, were eliminated from the analysis.

Data were summarized as percent of total time observed (%TTO) for duration behaviors and frequency/1000 sec (F/1000) for frequency behaviors. Means were calculated for each week of baseline, separation, and reunion, except for the first day of separation and the first day of reunion, which were treated separately.

Analyses of variance (ANOVA) were computed using a mixed factor design with species (bonnet, pigtail) and condition (control, separated) as between subject variables and phase [baseline week 1 (BW1), baseline week 2 (BW2), separation day 1 (SD1), separation week 1 (SW1), separation week 2 (SW2), reunion day 1 (RD1), reunion week 1 (RW1), and reunion week 2 (RW2)] treated as within subject variables. Separation day 1 and reunion day 1 were considered to be sufficiently different from the other days to be treated separately (see Reite, Short, Seiler, & Pauley, 1981). A total of 6 observation sessions were included on SD1 and RD1. SW1 and RW1 phases included a total of 8 sessions and SW2 and RW2 included 10 sessions. Simple effects analyses were conducted on significant interactions, followed by Neuman-Keuls tests for differences between individual means, and only significant results ($p < .05$) from these comparisons are discussed in the *Results* section.

Results

Behavioral Responses to Maternal Separation

A brief 2-week separation of an infant monkey from its mother while the infant remained in its natal social group was associated with a number of changes for both pigtail and bonnet monkeys. Not surprisingly, however, there were important species differences in the infants' response to separation. Observations of the control animals during the separation interval indicated that the removal of the adult female, and perhaps the resultant agitated behavior of the separated infant, were associated with transient disturbance in control animals as well. The following description has been divided according to the five categories in Table 3 for discussion.

Activity/Environmentally Directed Behaviors

Both pigtail and bonnet monkey infants showed changes in activity and environmentally directed behaviors as a function of the separation experience. Locomotion showed a significant phase by condition interaction ($F = 3.09$, 7/238 *df*, $p = .0039$), such that separated but not control bonnet and pigtail infants increased locomotor behavior on SD1. Rest showed a complementary decrease indicated by a phase by condition interaction ($F = 3.55$, 7/238 *df*, $p = .0012$), with reduced resting behavior on SD1, as well as during SW1 and SW2, and increased rest on RD1. Exploratory behavior showed a significant phase by species interaction ($F = 2.67$, 7/238 *df*, $p = .0113$). Separated pigtail infants decreased exploration

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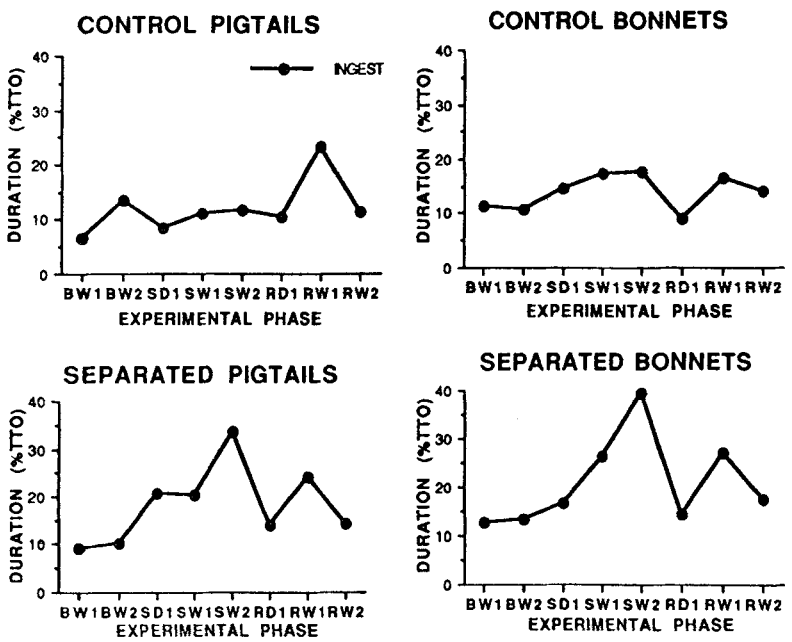


Fig. 1. Ingestive behavior as a function of separation and reunion for separated (lower panels) and matched control subjects (upper panels) for pigtail (lefthand panels) and bonnet (righthand panels) monkeys. (BW1, BW2 refer to the averages for baseline weeks 1 and 2, respectively; SW1, SW2 refer to separation weeks 1 and 2 respectively; RW1, RW2 refer to reunion weeks 1 and 2, respectively; SD1, RD1 refer to separation and reunion days 1, respectively; % TTO refers to percent of total time observed.) The same abbreviations are used for the remaining figures.

on SD1 and RD1, while separated bonnets showed no change in exploratory behavior.

Self-Maintenance Behaviors

Self maintenance behaviors were disrupted in both species by the separation experience. Ingestive behavior (eating and drinking) increased during the separation phase (phase by condition interaction $F = 3.5$, 7/238 *df*, $p = .0014$), in only the separated infants of both species (see Figure 1). Control subjects showed transient changes in ingestive behavior: a decrease on reunion day 1 in bonnet infants and an increase over reunion week 1 in pigtail infants. These rather complicated changes may be related to differences in group processes that reflect the removal of an adult female from the social group. The long-lived change in the pigtail infants may simply be a species difference. For both species, the separated subjects showed the greater change in ingestive behavior. Self-groom reflected a transient disruption, increasing slightly on SD1 (phase effect $F = 2.92$, 7/238 *df*, $p = .006$), but not on other days of the separation or at reunion. The magnitude of this effect was quite small, albeit significant. No species differences were noted in the self-maintenance category.

Disturbance Behaviors

Behaviors in the general category of disturbance (see Table 3), including vocalization, self-directed oral behavior, startle, shaking, or scratching, were

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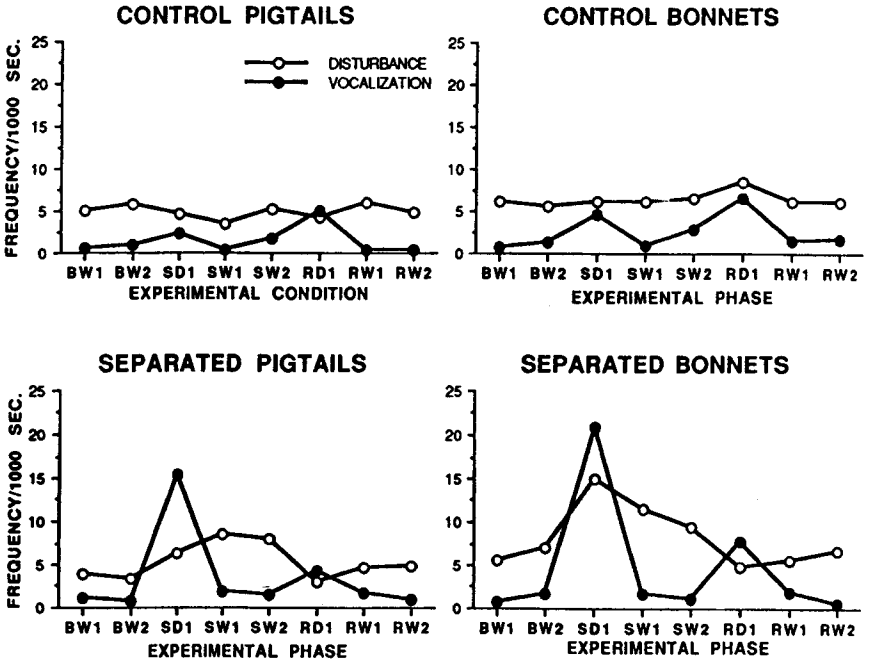


Fig. 2. Disturbance frequency behaviors as Frequency/1000 sec. See Figure 1 for details.

rarely observed under baseline conditions. They increased significantly in response to separation in both bonnet and pigtail separated infants as indicated in Figures 2 and 3. The responses of separated bonnet and pigtail infants were somewhat different: bonnet infants showed sustained agitated behavior (vocalization, shaking, temper tantrums), while, after an initial agitation period, pigtail infants became withdrawn, spending more time in withdrawn, slouched postures (Figure 3). Specifically, bonnet infants exhibited overall higher levels of behaviors in the disturbance category (species effect, $F = 16.60$, 1/34 *df*, $p = .0003$). Separated bonnet infants also showed a transient increase in both self-orality (on SD1: phase by condition by species interaction, $F = 2.46$, 7/238 *df*, $p = .0185$) and in stereotypic motor behavior (during the first week of separation: phase by condition by species interaction, $F = 2.09$, 7/238 *df*, $p = .045$). Separated bonnet and pigtail infants showed a dramatic increase in vocalizations on SD1 (phase by condition interaction, $F = 5.36$, 7/238 *df*, $p = .0000$). Separated pigtail infants showed slouched postures during the entire separation period (phase by species by condition interaction, $F = 7.75$, 7/238 *df*, $p = .0087$, Figure 3). Slouched postures were not observed in bonnet infants during the separation from the mother.

Nonmaternal Social Behaviors

Separated bonnet and pigtail infants showed different changes in their relationships with other group members, including agonistic behaviors, contact with group members, and allogrooming as discussed below. There were similar changes in some behaviors such as play, for which a significant phase by condition interaction was found ($F = 4.64$, 7/238 *df*, $p = .0001$). Play decreased during separation for

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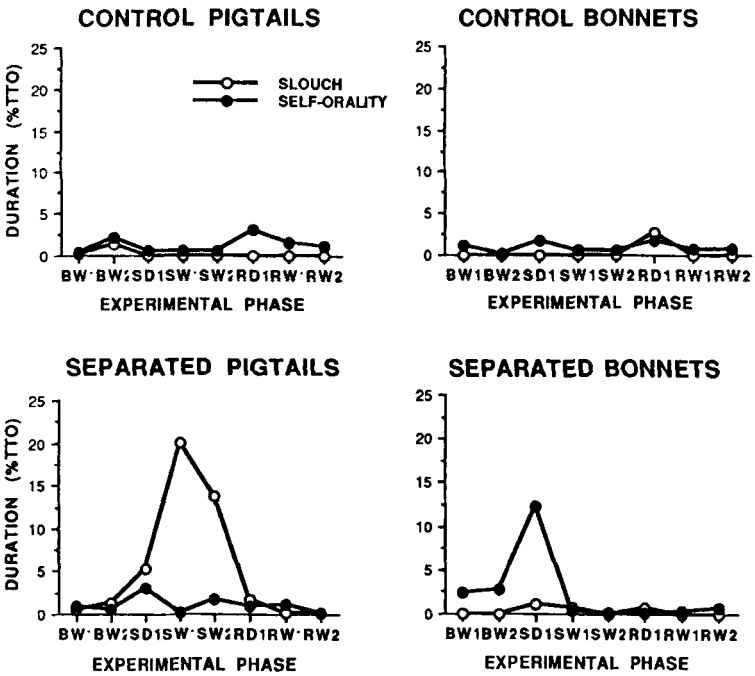


Fig. 3. Disturbance duration behaviors as % TTO. See Figure 1 for details.

all separated animals, although pigtail monkeys showed a greater decrease in play during separation than bonnet infants (phase by species interaction, $F = 2.94$, 7/238 *df*, $p = .0056$), as indicated in Figure 4.

Several social behaviors revealing the process of allomaternal care and adoption were noted in separated bonnet but not pigtail infants during maternal separation. Separated bonnet infants increased initiation of social groom on SD1 (phase by species interaction, $F = 2.08$, 7/238 *df*, $p = .046$). Separated bonnet infants also showed overall higher levels of agonistic behaviors, i.e., receiving more aggression and threats (Species $F = 10.64$, 1/34 *df*, $p = .0025$), as shown in Figure 5. This behavior increased during SW1 (phase by condition interaction, $F = 2.62$, 7/238 *df*, $p = .0126$), suggesting resistance on the part of some group members to the infant's attempts to facilitate social interactions. Yet, separated bonnet infants were successful in establishing ventral/ventral contact with group members during the separation resulting in a significant phase by species by condition interaction ($F = 3.15$, 7/238 *df*, $p = .0033$). This increase in contact with others was specifically related to the behavior of the group member in active adoption of the infant, and not the infant initiating proximity to the group member, since there was no increase in the mutually exclusive proximity category of near others, an infant initiated behavior.

Changes in behavior that appeared to be the result of the removal of an individual from the group and common to both control and separated infants were also noted. For example, an indication of potential group disruption was reflected in the reduction of play behavior in both controls and separated animals on the first days of separation and reunion (see Figure 4).

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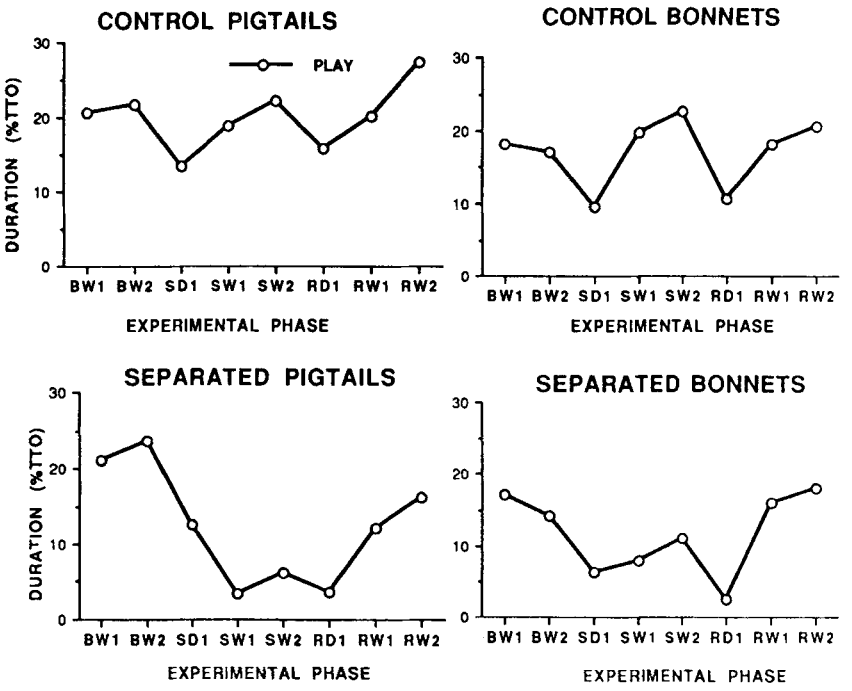


Fig. 4. Play behavior. See Figure 1 for details.

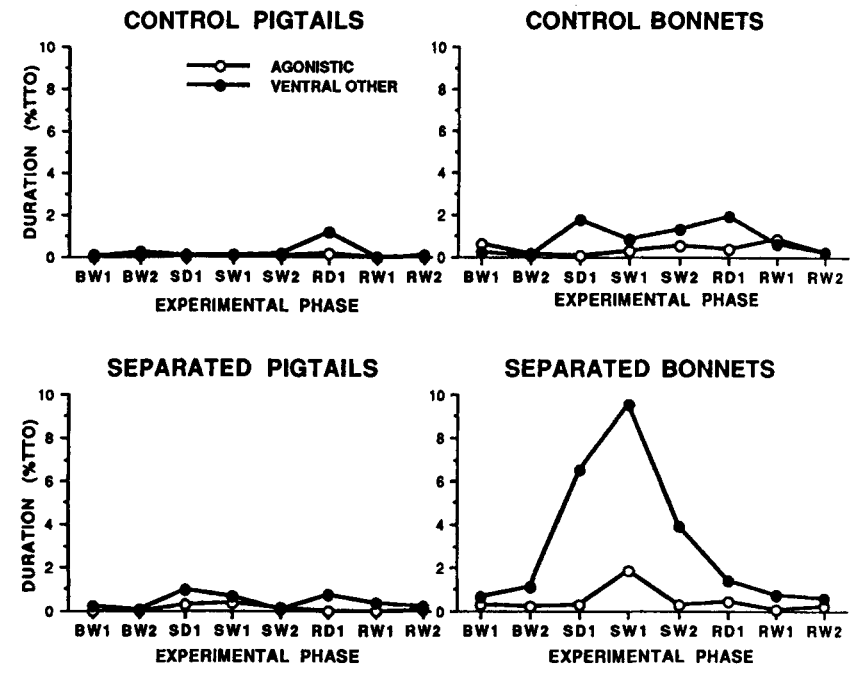


Fig. 5. Process of adoption is reflected by the increase in ventral other for the separated bonnet infants only, which was associated with an increase in agonistic behaviors directed toward the infant. See Figure 1 for details.

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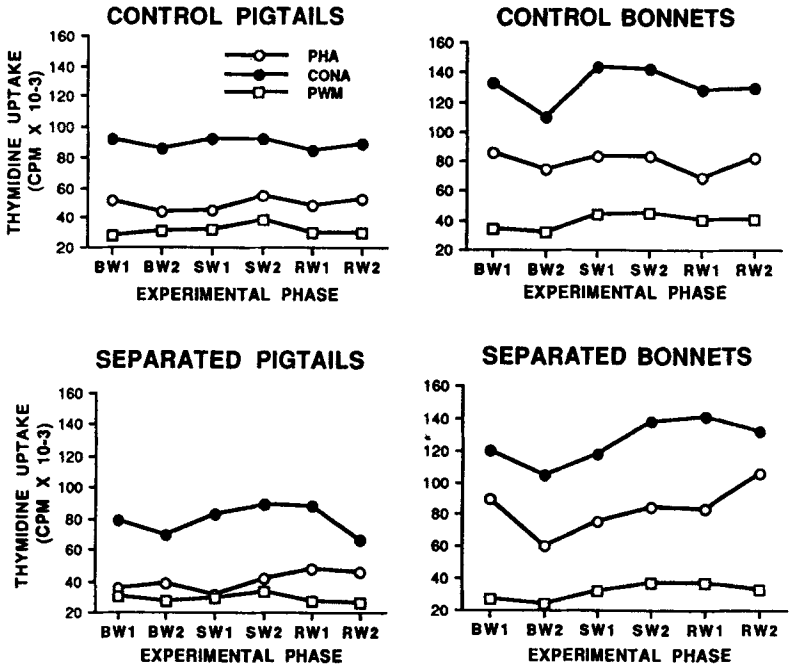


Fig. 6. Mitogen response for PHA, ConA, and PWM. The mean of the two weekly tests are given in the figure for the peak mitogen concentration as indicated in the text. See Figure 1 for details.

Maternal Social Behaviors

There were no significant changes in most maternal behaviors from baseline to reunion period, including ventral/ventral contact and weaning behavior in the separated infants. There were some differences, however, in affiliative behaviors, such as proximity. Affiliative behaviors (see Table 3) of separated bonnet infants were greater during both baseline and reunion periods (Species $F = 15.05$, 1/34 *df*, $p = .0005$); and on RD1 the previously separated bonnet infants were found in contact with their mothers more (phase by species interaction, $F = 2.55$, 4/136 *df*, $p = .0421$) than pigtail infants, albeit the magnitude of these changes was small.

Immunological Effects of Maternal Separation

There were significant species differences for the mitogen responses to both PHA ($F = 19.66$, 1/22, $p = .0002$) and ConA ($F = 16.11$, 1/22, $p = .0006$), but not to PWM for samples obtained during the baseline period. Lymphocytes obtained during the baseline period from both separated and control bonnet monkey infants showed greater proliferation in response to all concentrations of these mitogens than did separated and control pigtail monkey lymphocytes. These observations can be seen for BW1 and BW2 for the peak (identical for both species) mitogen concentrations in Figure 6. Total white blood cell counts and differential cell counts are given in Table 4 for both species. There were no significant species differences in either total white blood cell counts or differential cell counts for samples obtained during the baseline periods.

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Table 4
Mean Hematological Parameters for Infant Macaque Monkeys.

		Total WBC	% Lymphs	% PMN	% Mono
Pigtail	M	14,505	76.5	21.2	2.2
Macaques[1]	SD	5,263	11.5	11.1	1.2
Bonnet	M	16,602	76.5	21.2	2.0
Macaques[2]	SD	4,802	7.7	7.5	1.4

[1] N = 8 females, 7 males
[2] N = 7 females, 4 males

An ANOVA revealed that there were no significant effects of either experimental group (separated/control) or phase (baseline/separation/reunion) of the experiment on lymphocyte activation by the three mitogens. The dose response relationship between mitogen concentration and the proliferative responses of the lymphocytes as a function of phase of the experiment did not change during separation. In a few separation studies, subjects were bled 2 hours following maternal separation in an attempt to determine if there were any immediate effects of the separation on the immune panel. Results of the assays from blood drawn 2 hours after separation were unrelated to the experimental condition, although the number of subjects was small (2 bonnet pairs and 3 pigtail pairs). Separation was not associated with significant changes in the total white cell count or differential cell counts for samples obtained throughout the separation. These observations were unexpected, for it was predicted that there would be a phase effect that reflected the impact of separation on the proliferative response of experimental subjects. Lymphocyte activation responses following mitogen stimulation are shown in Figure 6 for the optimal concentrations of PHA (.25 ug/well), ConA (.75 ug/well), and PWM (3.0 ug/well) that produced the maximal proliferative response.

Subjects varied with regard to their behavioral responses to separation. Several specific behaviors were selected as covariates in an analysis of covariance (ANCOVA) to statistically control for the individual variation in individual responses to separation. These behaviors (covariates) included ventral-ventral contact with the mother during the baseline period (a measure of mother-infant attachment), time away from mother (a measure of infant independence), disturbance, vocalization on separation day 1, vocalization during separation, slouch on separation day 1, and slouch during separation. Lymphocyte activation by mitogens during separation weeks 1 and 2 were analyzed as the percent change from baseline observations.

Behaviors reflecting maternal interactions did not covary with changes in lymphocyte activation during the separation interval. When vocalization on separation day 1, slouch, and disturbance were treated as covariates, important relationships emerged from the analysis. A reduction in the mitogen response relative to baseline emerged during the second week of separation that was related to behavior. When slouch and vocalization were treated as covariates in the analysis, a significant effect of condition was found during the second week of separation for both T cell mitogens, PHA (F = 4.68, 1/18, p = .0442) and ConA (F = 6.59, 1/18, p = .0194). Slouch during separation week 1 significantly covaried with the change in the response to the peak mitogen concentration for both PHA (F =

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6.94, 1/18, $p = .0168$) and ConA ($F = 4.43$, 1/18, $p = .0497$). Vocalization on separation day 1 also significantly covaried with both PHA ($F = 5.15$, 1/18, $p = .0357$) and ConA ($F = 5.88$, 1/18, $p = .026$) responses. For both covariates, as the particular behavior increased, lymphocyte activation by the T cell mitogens decreased. Covariance analysis showed that the B cell mitogen, PWM, was not significantly related to experimental phase, vocalization, slouched postures, or disturbance. Interestingly, there was a *positive* relationship, albeit nonsignificant, between disturbance and the response to PWM for both separation weeks 1 ($r = +.57$) and 2 ($r = +.66$). In other words, the higher the disturbance score, the greater the increase in the PWM response during both the first and second week of separation, relative to the baseline period. There was no effect of species in this analysis.

To summarize the present observations, monitoring the affective consequences of maternal separation is important in the prediction of changes in lymphocyte activation by T cell mitogens from baseline to the second week of separation. In the present analysis, the particular species was not important, albeit the pigtail monkeys were the only subjects that showed slouched, withdrawn postures; rather it was the magnitude of the components of the behavioral response associated with separation that discriminated the subjects that revealed an alteration in lymphocyte function.

Discussion

The present study provides important observations regarding behavior and immune relationships during maternal separation in nonhuman primates, in addition to further describing species differences in the behavioral response to maternal separation and the possible role of social support in this process. Immunological consequences of separation covaried with several behavioral responses associated with the separation experience, namely vocalization on the first day of separation and time spent in withdrawn slouched postures during the first week of separation. The behavioral responses will be discussed briefly, followed by a discussion of the implications of the immunological observations.

Three distinct behavioral processes were noted in response to separation of an infant macaque from its mother. For both species, the initial response was one of agitation and protest which was characterized by increased vocalization, activity, and ingestive behaviors. A second phase, observed only in pigtail infants, followed the agitation phase. This depressive phase followed within 24–48 hours of separation, during which time the separated pigtail infants were less likely to engage in play behavior and were seen in withdrawn, slouched postures. The failure to observe a similar phase in separated bonnet infants may be related to the development of allomaternal care in this species, reflected by an increase in the amount of time the separated infant spent in contact with other group members (see Figure 5). In the absence of resources for allomaternal care, the bonnet infant will show a prominent depressive phase (Kaufman & Stynes, 1978; Reite & Snyder, 1982). A process of disturbance was noted in both the control and separated infants of both species on the first day of separation and reunion. This may be related to

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the removal and return of an adult animal to the social group and/or the agitated behavior of the separated infant.

These responses were not seen uniformly across all subjects. Some pigtail infants failed to show postural collapse, for example, whereas others spent as much as 50% of the observation interval in that posture or some other expression of disturbed behavior. The removal of the mother from the social group is not the only event that precipitates the depressive phase. We noted, anecdotically, that rejection of pigtail infants by the mother after the separation experience was also associated with a depressive response by the infant. On the first day of reunion, the mother was reestablishing her relationship with the alpha male and other adult females in the group while the infant was ignored or actively pushed away by its mother. Bonnet infants failed to show this response, appearing content to remain with other group members during early phases of reunion.

Not all of the changes observed in the separated infants were related to maternal separation. Changes in play behavior occurred in both separated and control subjects, and as a primary affiliative behavior, changes in play are an important indicator of disturbance. The changes in play behavior of the controls were specific to the first day of separation and reunion and are not likely to represent the impact of blood sampling, as the change was specific to those days in only control subjects.

The process of the development of allomaternal care in the bonnet infants is of interest as a potential model of social support. This process would seem to be unique to the bonnet infants. There was no indication of a similar process in the pigtail infants, which showed increasing levels of disturbance, slouched postures, and no recovery in play until after reunion. Agitation in bonnet infants was greatest during the first week of separation, during which time the infant received the highest level of aggression from members of the social group. The infants actively sought contact with other group members during this time (see Figure 5), for which there was an increase in agonistic behavior directed toward the infant. By week 2 of separation, disturbance behavior had declined in the separated bonnet infants as had the agonistic behaviors directed toward the infants. The process of adoption, however, did not appear to have a direct impact on the immunological changes.

A major contribution of this study is the demonstration of the relationship between a stressor, its behavioral correlates, and changes in a measure of immune function, lymphocyte activation. Such associations do not necessarily reflect causation and must be interpreted carefully. The covariance of behavioral changes associated with maternal separation and changes in T cell activation by mitogens could simply be epiphenomena of a more basic process occurring during separation. Similar relationships have been demonstrated in another animal model with different immune markers (see Maier & Laudenslager, 1988). In adult male rats, the experience of territorial intrusion is associated with reduced plasma levels of specific antibodies to a protein administered at the time of group introduction (Fleshner, Laudenslager, Simons, & Maier,1989). If intruders were categorized as defeated or undefeated by median split of the amount of time spent in submissive postures, 46% of the variance in antibody level was accounted for by the group to which the subject belonged.

In the present study, there were no differences in lymphocyte activation between separated and control groups as indicated in Figure 6. The covariance

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analysis indicated that the change in lymphocyte activation by T cell mitogens from baseline to week two of separation covaried with the amount of time spent in slouched, withdrawn postures during the first week of separation and vocalization on the first day of separation. These observations would seem to parallel those in defeated rodents; that is, it is the behavioral correlates of the stressor and not the specific stressor that are important in predicting immune changes. The observations presented in Figure 6 would seem to be inconsistent with prior observations from this laboratory (Reite, Harbeck, & Hoffman, 1981; Laudenslager, Harbeck, & Reite, 1982). These earlier studies reported reduced lymphocyte activation by mitogens associated with either peer or maternal separation. Further studies of lymphocyte activation by mitogens during peer separation revealed a significant reduction only in the response to pokeweed mitogen, a B cell mitogen, during the second week of separation (Boccia, Reite, Kaemingk, Held, & Laudenslager, 1989). There were no significant changes in the response to the T cell mitogens associated with peer separation. Covariance analysis was not applied in that study due to the small sample size. As stated earlier, it is important to describe behavioral responses to separation in order to predict immunological consequences. Indeed, this has been emphasized in studies in human populations (Kobasa, 1979; Wortman & Silver, 1987).

Does brief maternal separation in nonhuman primates serve as an adequate animal model for loss and bereavement? There are a number of similarities between maternal separation in nonhuman primates and the process of bereavement in humans (Laudenslager, 1988). The increased incidence of morbidity and mortality in the recently bereaved (Stroebe & Stroebe, 1987) requires that some underlying change in biological function occurs that might place the individual at risk. The changes in autonomic function (Reite, Short, Seiler, & Pauley, 1981) and immune regulation as indicated here and elsewhere (Coe, Rosenberg, Fischer, & Levine, 1987; Coe, Rosenberg, & Levine, 1988; Laudenslager, Harbeck & Reite, 1982; Laudenslager, Reite, & Held, 1986) would seem to provide a number of possible mechanisms whereby health might be impaired. A growing literature suggests that major affective disorders are associated with changes in immune regulation (Stein, 1989) but in a rather complicated manner. For example, age determines if major depressive disorders are associated with immunological changes. Older individuals were more likely to show reduced lymphocyte activation than younger subjects (Schleifer, Keller, Bond, Cohen & Stein, 1989).

The relationship of *in vitro* lymphocyte activation to health is far from clear (Maier & Laudenslager, 1988; Stein, 1989). As a multifaceted system with many backups and regulators, changes of an *in vitro* measure associated with an affective state must be considered cautiously with regard to the biologic implications. Changes in the production of specific antibodies following challenge with a novel antigen is reflective of the *in vivo* response of the immune system (see Maier & Laudenslager, 1988). A pilot study from our laboratory indicated that the ability to produce specific antibodies to a foreign protein was reduced in pigtail infants undergoing maternal separation experience (Laudenslager, Reite, & Held, 1986). We have replicated these observations in a larger sample of both pigtail and bonnet infants.

The mechanism whereby the alterations in immune function are effected is far from established. Maternal separation in nonhuman primates is accompanied by

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activation of the hypothalamic-pituitary-adrenal (HPA) axis (Levine, Johnson, & Gonzalez, 1985; Smotherman Hunt, McGinnis, & Levine, 1977) with an associated increase in circulating corticosteroids, catecholamines, and endogenous opiates, all of which have immunomodulatory effects (Bateman, Singh, Kral, & Solomon, 1989). Changes in HPA regulation in major affective disorders (Kathol, Jaeckle, Lopez, & Meller, 1989) could account for changes in immunoregulation noted in populations of depressed patients. We did not monitor plasma glucocorticoids in the present study because of the time required to obtain the blood sample. We have developed new procedures that now permit rapid sampling of both separated and control infants in less than 10 minutes. This will provide a better assessment of basal plasma cortisol levels.

In summary, brief maternal separation experiences are associated with species dependent changes in the behavior of the separated infant. Bonnet infants show an agitation phase that is terminated with an increase in allomaternal care, whereas in pigtail infants, in the absence of allomaternal care, the agitation phase progresses to a depressive phase. Both species show similar disruptions in self-maintenance behaviors and play behavior. In both species the immunological changes were significantly related to the behavioral responses associated with separation.

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